Ex1 - Command Line & P0 - My First Kernel Mod

COP4600

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Assignment Philosophy

- OS Exercises and Projects will not explicitly hold your hand
- You will be given tasks to complete and formatting to follow, but you will not always be told exactly *how* to complete those tasks within the assignment doc itself.
- Resources will typically be provided to you in the form of links to documentation or relevant articles.
- Generally, assignments will heavily hint as to how you would complete them, but it is expected and encouraged that you look up other resources yourself!
- If you ever feel completely lost, please ask for help (early)!

Ex1 - Command Line

- The main way you will be interacting with Reptilian throughout this course is through your SSH'd terminal, not the GUI.
- Ex1 will let you get familiar with the terminal and terminal commands.
- Command 'flags' are options that alter command behavior.
- For example: typing 'ls' into your terminal will list all the files in your current directory. Typing 'ls -S' will list all the files in your current directory sorted by file size.
- Please get familiar with all the commands on the cheat sheet at the bottom of the Ex1 doc, but we'll show you a few important ones next.

Ex1 - Command Line (cont.)

- find
  - Used to search for certain using their properties: filename, file extension, file size, file permissions, etc.
  - Not for searching for specific file contents.
  - Example:
  - find /home/reptilian -name "file.txt"
  - This searches /home/reptilian for a file with a name that contains “file.txt”.

For further info.

Ex1 - Command Line (cont.)

- grep
  - Used to search for patterns inside file contents.
  - Not for searching through file names or attributes.
  - Example:
  - grep -R “phrase” /home/reptilian
  - This will recursively (-R) search the /home/reptilian directory for files containing the text “phrase” in their file contents.

For further info.

Ex1 - Command Line (cont.)

- (Go over pdf and demo terminal functionality, like navigating directories, viewing files, and zipping/tarring files.)

Man files and Project reports

- ‘Man’ is a markup language used for text formatting on man pages.
- Excluding Project 0, all future project reports must have man formatting for full credit. It is recommended that you also submit P0 in man format, but it is not required.
- Project man files will be submitted as .txt files, and the markup language must be visible.
- Format your reports nicely. No big blobs of text/huge paragraphs please.
- (Show man_page.txt example on canvas.)
- Helpful tool for editing man pages online

Report content - Technical Details

- List the complete filepaths of any files you altered, and any files containing relevant source code for your project. Explain what those files are, and why you needed to create/change them.
- Explain your approach to the project. What data structures did you use? How are things maintained and managed? What led you to make those decisions?
- How did you complete the project? List commands you ran, why you ran them, and explain the meaning behind their flags if they are significant.
- We want to see EVERY function mentioned, even getters and setters.
- Reports must be your own. Plagarism is serious.
- Projects will posted files specifically detailing what we want in each report.

Report content - Testing

- How did you test your project?
- Did you use some provided testing suite we gave you?
- Did you test any kernel changes by rebuilding the kernel and rebooting the machine?
- Your implementation MUST be tested on a clean kernel (snapshot from Ex0)! We need to make sure your solution generalizes to any machine with the OS.
- All of this is important to mention!

Project 0 - My First Kernel Mod

- It is recommended that you complete Ex1 before starting P0, as commands you learn in Ex1 will prove useful to you in starting P0.
- The assignment doc heavily hints at the location where you need to add the message, do your best to find the correct file and place!
- Near the call where you should place the message is a good hint as to how to get a KERN_ERR to print… look at how the kernel code already does it!
- Take snapshots frequently when making kernel mods. You do not want to brick your kernel and lose progress because you didn't back any of it up.
- (Go over project doc, and show what the message and extra credit message should look like on the startup screen)

Project 0 - My First Kernel Mod (cont.)

- Patch files must be applied to a clean kernel. (Ex0 Kernel Snapshot)
- Your screencast will likely be retreading material from your report. That is okay, they are graded independently of each other.
- You are heavily encouraged to explain how you arrived at your solution, what commands you needed to run, and show the code where you added the line with the message.
- This applies to all future screencasts too; please show all code, and verbally walk through each function, and demonstrate testing.
- All Changes -> Main Functionality -> Testing!
- You can use a video editor to cut out boring parts of a recording (long, slow boot sequence), but no speed up! Remember, you're on a time limit.